

# CS105 (DIC on Discrete Structures)

## Exercise problem set 13

- Attempt *all* questions.
  - Apart from things proved in lecture, you cannot assume anything as “obvious”. Either quote previously proved results or provide clear justification for each statement.
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### Part 1

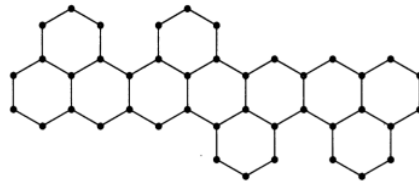
1. Give an example of the stable matching problem with two men and two women in which there is more than one stable matching.
2. Determine the stable matchings resulting from the Proposal Algorithm run with men proposing and with women proposing, given the preference lists below.

**Men**  $\{u, v, w, x, y, z\}$

**Women**  $\{a, b, c, d, e, f\}$

|       |                         |       |                         |
|-------|-------------------------|-------|-------------------------|
| $u :$ | $a > b > d > c > f > e$ | $a :$ | $z > x > y > u > v > w$ |
| $v :$ | $a > b > c > f > e > d$ | $b :$ | $y > z > w > x > v > u$ |
| $w :$ | $c > b > d > a > f > e$ | $c :$ | $v > x > w > y > u > z$ |
| $x :$ | $c > a > d > b > e > f$ | $d :$ | $w > y > u > x > z > v$ |
| $y :$ | $c > d > a > b > f > e$ | $e :$ | $u > v > x > w > y > z$ |
| $z :$ | $d > e > f > c > b > a$ | $f :$ | $u > w > x > v > z > y$ |

3. Exhibit a perfect matching in the graph below or give a short proof that it has none.



## Part 2

4. A deck of  $mn$  cards with  $m$  values and  $n$  suits consists of one card of each value in each suit. The cards are dealt into an  $n \times m$  array.
  - (a) Prove that there is a set of  $m$  cards, one in each column, having distinct values.
  - (b) Use part (a) to prove that by a sequence of exchanges of cards of the same value the cards can be rearranged so that each column consists of  $n$  cards of distinct suits.
5. Use the König–Egerváry Theorem to prove that every bipartite graph  $G$ , where  $e(G)$  denotes the number of edges in  $G$  and  $\Delta(G)$  denotes the maximum degree of  $G$ , has a matching of size at least

$$\frac{e(G)}{\Delta(G)}.$$

Use this result to conclude that every subgraph of  $K_{r,r}$ , the complete bipartite graph with  $r$  vertices in each part, and with more than  $(k-1)r$  edges, has a matching of size at least  $k$ .

6. Prove that if man  $x$  is paired with woman  $a$  in some stable matching, then  $a$  does not reject  $x$  in the Gale–Shapley Proposal Algorithm with men proposing. Conclude that among all stable matchings, every man is happiest in the matching produced by this algorithm. (*Hint:* Consider the first occurrence of such a rejection.)
7. In the *Stable Roommates Problem*, each of  $2n$  people has a preference ordering on the other  $2n-1$  people. A *stable matching* is a perfect matching such that no unmatched pair prefers each other to their current roommates. Prove that there is no stable matching when the preferences are those below.

$$\begin{aligned} a : & \quad b > c > d \\ b : & \quad c > a > d \\ c : & \quad a > b > d \\ d : & \quad a > b > c \end{aligned}$$